

PORTFOLIO RESUME

Ayush Maurya

Computer Science Engineering student and researcher at IIIT Hyderabad working at the intersection of computer science and human sciences.

Software Engineering
Machine Learning
Human Sciences

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ABOUT

2021 Started B.Tech/M.S. 111 UGEE Rank

Dual-degree student at IIIT Hyderabad combining a B.Tech in Computer Science Engineering with an ongoing M.S. by Research in Computing and Human Sciences. Interests include machine learning, software systems, and human-centered technology.

EXPERIENCE

SERC Digital Twin Environmental IoT
Built a real-time environmental monitoring dashboard in MERN by integrating local temperature, humidity, water, and air quality sensors.
MERN • IoT • SQL

TAFEA System Design
Designed a student client software system using SDLC methods and LLM APIs for tailored student-facing features.
System Design • UI Design • LLM

SELECTED PROJECTS

ML for Local Geoguesser Python • ML • PyTorch
Built a ResNet50-based location prediction model for latitude, longitude, region, and angle, reaching 92% region accuracy.

ElectoralSim Python • Simulation • Open Source
Developed an open-source Python library for simulating electoral systems and voting methods as part of human sciences research.

CrowdTwin Next.js • Product Design • Prototype
Created a predictive crowd monitoring dashboard prototype for a tech product entrepreneurship course.

Design for Social Innovation (TAFEA) Human-Centered Design • Education • AI
Designed an AI-assisted platform to help Teach For India fellows plan extracurricular curricula for resource-constrained classrooms.

Highland History Project Machine Learning • OCR • LLM
Used OCR and LLM-assisted text recovery to digitize historical documents for Himalayan trade network research.

EDUCATION

IIIT Hyderabad 2021 - Present
B.Tech in Computer Science Engineering (CSE)
M.S. by Research in Computing and Human Sciences (Ongoing)

Relevant Coursework & Labs 2022 - 2024
Design & Analysis of Software Systems
Human Science Lab (NLP Research)

Mary Gardiner's Convent School 2021
Class XII - 90.1%

City Montessori School 2019
Class X - 94.2%

EXPERTISE

- **Languages:** Python, C/C++, JavaScript, SQL, HTML/CSS
- **Technologies:** Machine Learning, PyTorch, Linux, System Design, Systems Programming
- **UI/UX & Design:** Wireframing, Prototyping, User Stories, User Research, Interaction Design

HONOURS

UGEE: Rank 111 out of 40,000+ candidates • JEE Main: 96.9 percentile • Activities: Bass guitarist, lawn tennis

ADDITIONAL LINKS

Research • IPL Strategy Lab • TempusLogic • WebTTS • amReader • IIIT In Context • Mental Health Toolkit • ZenFocus